INCH-POUND

MIL-I-3190/9
8 August 1991

MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, FLEXIBLE, SILICONE COATED CLASS 240°C, TYPE D, CATEGORY C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the insulation sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solication: MIL-I-3190

The sleeving shall be of the following classification:

Class 240°C, Type D, Category C

REQUIREMENTS:

The insulation sleeving shall be in accordance with tables I, II, and III.

TABLE I. Performance requirements

Property	Conditioning prior to test	Test	Unit	Value required
Dielectric breakdown:				
Straight	96/23/50	4.7.2.2	Volts (mininum average)	8,000
Straight	96/23/50	4.7.2.2	Volts (minimum individual)	6,000
Straight	96/23/96	4.7.2.2	Percent of average value obtained on test after condition 96/23/50 (minimum)	80
Cold brittleness	96/23/50	4.7.4	Degrees Celsius (maximum)	minus 70
Flammability	96/23/50	4.7.5.1 Method B	Seconds to burn 1" for each specimen minimum	45

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TABLE I. Performance requirements (Continued)

Property	Conditioning prior to test	Test	Unit	Value required
Push-back after heat aging:	168/250 1/	4.78	Observation	No cracks or rupture in the sil rbr on two or more specimens
Visual				
Dielectric breakdown: Straight	168/250 1/	4.7.8	Volts (minimum average)	6,000
Thermal endurance	see 4.7.9 1/	4.7.9	Temp index for (C) at 15000 hrs minimum	240
Thermal stability	96/250	4.7.10	Volts (minimum) average)	5,500
Hydrolytic stability	336 hours (14 days) over water at 70° C	4.7.7	Volts (minumum average) and observation	5000 No disin-tegration reversion or cracks

1/ Denotes increase in temperature resistance thus increase in test temperature.

TABLE II. Qualification inspection requirements.

Examination or test	Requirement	Test	Minimum number of tests for each size and for each condition as specified in table I
Visual examination	3.3, 3.6	4.6	10
Dimensions	3.4	4.7.1	10
Dielectric breakdown: Straight - 96/23/50 and 96/23/96	Table I	4.7.2.2	10
Cold brittleness	Table I	4.7.4	As specified in 4.7.4 (10)
Flammability	Table I	4.7.5.2 Method B	5

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TABLE II. Qualification inspection requirements (Continued).

Examination or test	Requirement	Test	Minimum number of tests for each size and for each condition as specified in table I
Push-back after heat aging	Table I	4.7.8	3
Thermal endurance	Table I	4.7.9	See 4.7.9
Hydrolytic stability	Table I	4.7.7	5

TABLE III. Quality conformance inspection requirements.

Examination or test	Requirement	Test	Conditioning prior to test	Minimum number of tests per sample unit
Visual examination	3.3, 3.6	4.6	96/23/50	As required per sampling
Dimensions	3.4	4.7.1	96/23/50	As required per sampling
Dielectric breakdown:				
Straight	Table I	4.7.2.2	96/23/50 96/23/96	10 10
Flammability	Table I	4.7.5.2 Method B	96/23/50	5
Push-back after heat aging	Table I	4.7.8	168/250	3
Thermal stability	Table I	4.7.10	96/250	5

Intended Use: The temperature index for this class of sleeving at 15,000 hours is 240°C. The sleeving covered by this specification is intended for applications in electrical equipment requiring primary insulation having good low temperature flexibility, high moisture and humidity resistance, excellent mechanical protection and abrasion resistance.

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In applications requiring resistance to the transfer of high molecular structures from the sleeving to insulating oils, a polyester varnish overcoat improves performance. This composite also exhibits ease of cleanup when carbon or graphite particles are prevalent. In addition to meeting the requirements of Table I, it should also be tested to 4.7.6.4.

In applications requiring good adhesion of silicons, tetrafluoroethylene fluorocarbon polymers or other materials to the sleeving, an investigation should be conducted to determine acceptability

Preparing activity:
Navy - SH
(Project 5970-1091)

Custodians:
Army - ER
Navy - SH
Air Force - 11
Review activities:
Army - AR, AV, EA, MI
Navy - EC
Air Force - 85, 99, 80
User activities:
Army - ME
Navy - MC, OS

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